12EC35 / 12TE35

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-15

Common to EC / TE

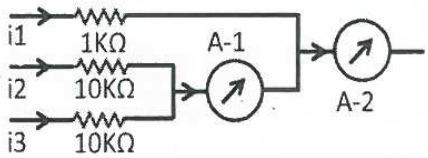
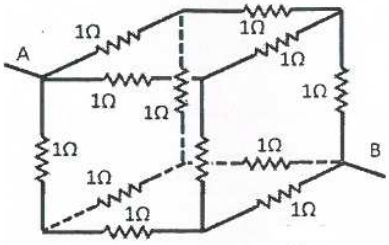
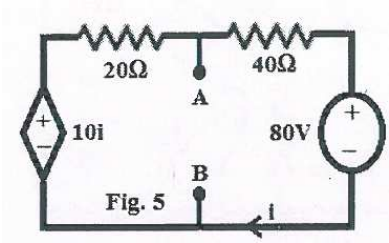
NETWORK ANALYSIS AND CONTROL THEORY

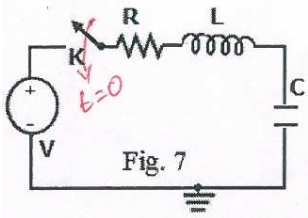
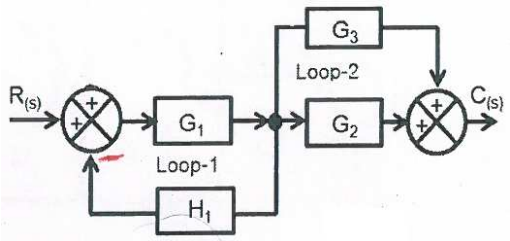
Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

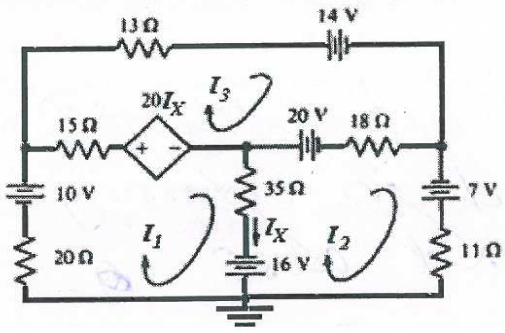
5. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
6. Answer FIVE full questions from Part B.

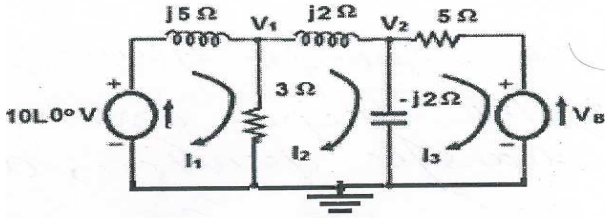
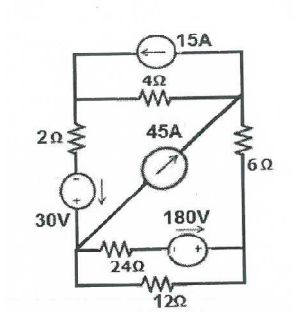
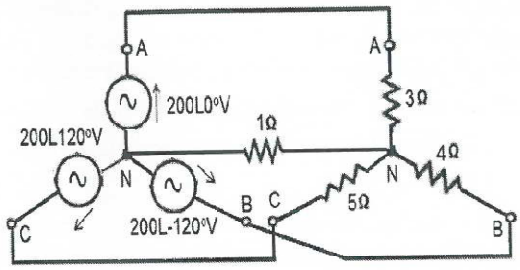
PART-A

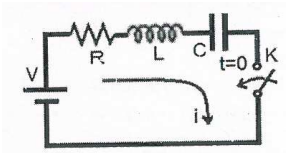
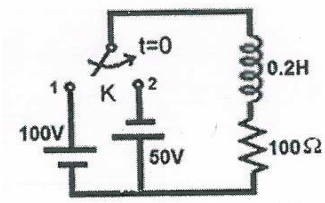
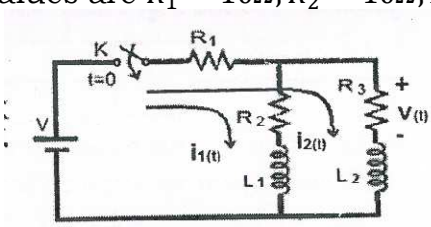
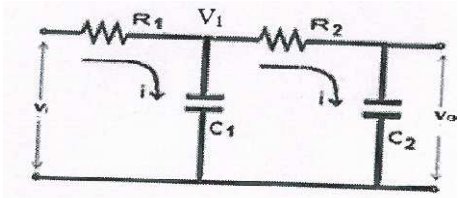
1	1.1	In the circuit of Fig 1.1, if $i_1 = 14.14 sin(\omega t + 45°)$, $i_2 = 14.14\ sin\ (\omega t - 75°)$ and $i_3 = 14.14\ Sin(\omega t - 195°)$, then find the reading on ammeter $A - 2$.  Fig 1.1	02
	1.2	For the arrangement shown in Fig 1.2, calculate the equivalent resistance between the diagonally opposite ends A and B.  Fig 1.2	02
	1.3	In a system transfer function, the poles and zeros, if complex, must be in _____.	01
	1.4	The load impedance Z_L connected to a generator with impedance $(10 - j10)\Omega$ has both R_L and X_L variables. To minimize the power loss from generator to load, suggest suitable value of Z_L. What is the power delivered to load, if the generator voltage is $10V$?	02
	1.5	For the network in Fig 1.5, the Thevinin's voltage across AB is $V_{AB} =$ _____.  Fig 1.5	01

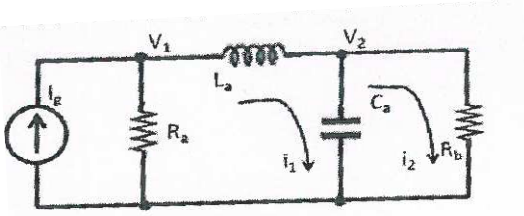
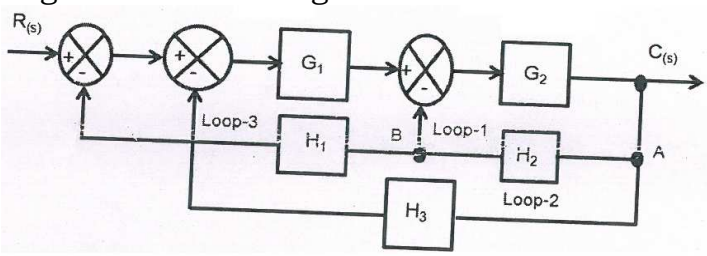
1.6	In the analysis of networks using Thevinin's and Norton's theorems, while estimating the impedance between the output terminals (Z_{Th}/Z_N), the sources in the network are _____.	01
1.7	In the circuit of Fig 1.7, all initial conditions are zero; switch K is closed at $t = 0$, then find $i(0+)$ and $\frac{d}{dt}i(0+)$, if $V = 10V$ and $L = 1H$.  Fig 1.7	02
1.8	A $10\mu F$ capacitor has a charge of $10000nC$ at $t = t_1$. Switching occurs at $t = t_2$, draw the equivalent for the capacitor at $t = \infty$.	01
1.9	Find the initial value and final value for the function $\frac{1}{s(s^2-a^2)}$.	01
1.10	For the block shown in Fig 1.10, the system transfer function $TF = C(s)/R(s)$ is _____. If all the path gains are reduced to unity, then the TF becomes _____.  Fig 1.10	02
1.11	Obtain the transfer function for RC integrator circuit.	01
1.12	Find centroid and the point of intersection of root locus on the imaginary axis for a unity feedback system with $G(s) = \frac{K(s+6)}{s(s+1)(s+2)}$.	02
1.13	The expression for steady state error of a first order system to a step input is _____ and to a unit ramp input is _____.	02

PART-B

2	a	Classify networks based on the network components. Give examples.	04
	b	Obtain the circuit for the set of equations given below: $7I_1 - 6I_2 + 12I_3 = 10$, $2I_1 - 12I_2 + 4I_3 = 0$ and $I_1 - I_3 = 5$.	04
	c	Using mesh analysis, find the power absorbed by the dependent voltage source in the circuit of Fig 2 (c).  Fig 2(c)	08

		OR	
3	a	Explain the following terms with examples: i) Super mesh ii) Super Node.	04
	b	Obtain the circuit for the set of equations given below: $$\begin{bmatrix} 7+j3 & -j5 & -5 \\ -j5 & 12+j3 & -(2-j2) \\ -5 & -(2-j2) & 17+j2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10\angle 0^\circ \\ -5\angle 30^\circ \\ -10\angle 90^\circ \end{bmatrix}$$	04
	c	For the network shown in fig 3(c), using Node analysis, find the value of V_B such that current through the capacitive reactance $-j2\Omega$ is only I_2.  fig 3(c)	08
4	a	With suitable examples, explain the principle of duality in networks.	04
	b	State and explain Thevenin's theorem.	04
	c	For the network shown in Fig 4(c), find the value of maximum power transferred to the 12Ω resistor.  Fig. 4(c)	08
		OR	
5	a	What is meant by dual networks? Explain the use of duality principle.	04
	b	State and explain Superposition theorem.	04
	c	Find the current flowing through and the power dissipated in the 1Ω resistor in fig 5(c) using Millman's theorem.  Fig. 5(c)	08

6	a	State and prove initial value and final value theorems.	08
	b	In the circuit shown in fig 6(b), switch is closed at $t = 0$. Obtain the value of $i(t)$ using Laplace transformations; given $R = 9\Omega, L = 2H$ and $V = 4V$ and $C = \frac{1}{7}F$. Assume zero initial conditions.  Fig. 6(b)	08
		OR	
7	a	In the circuit of Fig .7(a), switch is moved to position '1' at $t = 0$, there by applying $100V$ source to $R - L$ branch. At time $t = 500\mu s$, the switch is moved to position '2'. Obtain the equations for the current in both positions of the switch and sketch the waveforms of the transients.  Fig. 7(a)	08
	b	For the circuit shown in Fig .7(b), a step voltage of $V = 100V$ is applied by closing the switch at $t = 0$. Assuming no initial conditions, obtain the value of $v(t)$ across R_1 resistor using Laplace transformations. The circuit component values are $R_1 = 10\Omega, R_2 = 10\Omega, L_1 = 1H$ and $L_2 = 2H$.  Fig. 7(b)	08
8	a	What are the advantages and disadvantages of open loop control systems?	04
	b	For the system represented by the following set of equations, draw the signal flow graph; here U_1 and U_2 are input variables and X_1 and X_2 and X_3 are output variables. $X_1 = a_{11}X_1 + a_{12}X_2 + a_{13}X_3 + b_1U_1;$ $X_2 = a_{21}X_1 + a_{22}X_2 + a_{23}X_3 + b_2U_3;$ $X_3 = a_{31}X_1 + a_{32}X_2 + a_{33}X_3;$	04
	c	Construct the signal flow graph for the system shown in fig 8(c) and obtain the transfer function using Mason's Gain Formula.  Fig. 8(c)	08

		OR	
9	a	What are the advantages and disadvantages of closed loop control systems?	04
	b	For the network shown in fig 9 (b), obtain the block diagram.  Fig. 9(b)	04
	c	For the block diagram shown in fig 9 (c), obtain the transfer function using block diagram reduction algebra.  fig 9 (c)	08
10	a	What are the general characteristics of root locus diagrams?	04
	b	What are the steady state error constants? Explain briefly.	04
	c	Find K_p, K_V, K_a for a system with $G(s)H(s) = \frac{(s+10)}{s(s^3+7s^2+12s)}$. Also, evaluate e_{SS} when the input, $f(t) = 2tu(t)$.	08
		OR	
11	a	What are the time domain specifications of a second order system with a step input? Explain briefly.	04
	b	Sketch the RLD for a system whose open loop transfer function is given by $G(s)H(s) = \frac{K}{(s+2)(s^2+4s+3)}$. Also, find the value of K for which the system operates with a damping ratio of $\zeta = 0.5$.	12